EXPERIMENT No 8: Blinking LED with MIXLY

Student Guide

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Objective:

In this experiment, you will learn how to make an LED blink using MIXLY, a block-based programming environment for Arduino. This will introduce you to the basics of controlling electronic components with code and how to create simple time-based actions using Mixly blocks.

Materials Needed:

- Arduino Uno
- 1 LED
- 1 Resistor (220Ω or 330Ω)
- Breadboard
- Jumper Wires
- USB Cable (for connecting Arduino to your computer)
- Computer with Mixly software installed

What is MIXLY?

MIXLY is a visual programming environment that allows you to create code for Arduino by dragging and dropping blocks, similar to Scratch. It's an easy way to get started with programming hardware components without writing text-based code.

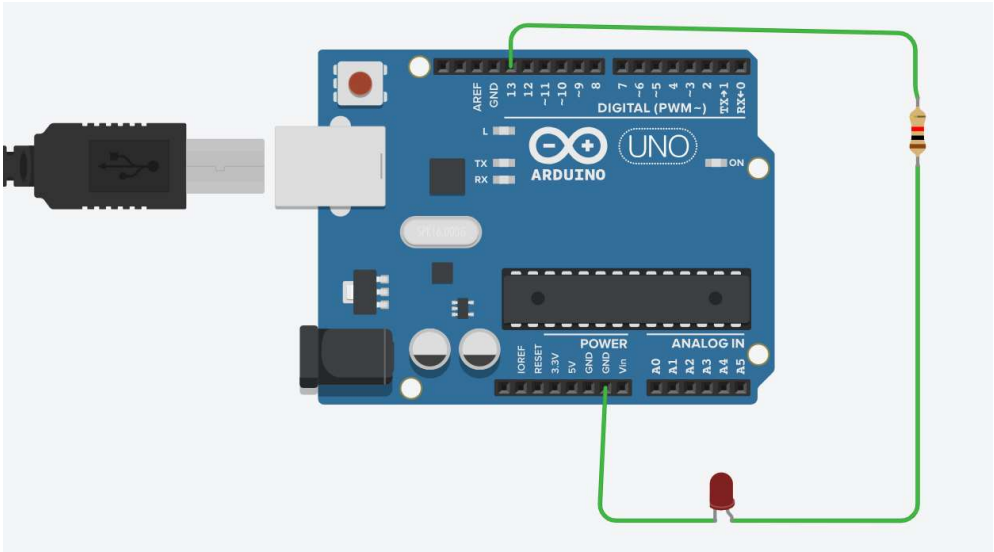
Steps to Blink an LED Using MIXLY

Step 1: Set Up Your Circuit

1. Connect the LED:
   - Place the LED on the breadboard. Make sure the longer leg (anode) is connected to digital pin 13 on the Arduino.
   - Connect the short leg (cathode) of the LED to the ground (GND) pin of the Arduino using a resistor in series to limit the current.

Circuit Diagram:

- **Pin 13 → Resistor → LED (long leg) → LED (short leg) → GND**



2. **Connect the Arduino to Your Computer:**
   - Use the USB cable to connect the Arduino Uno to your computer.

Step 2: Open Mixly and Set Up the Program

1. **Open Mixly:**
   - Start the Mixly program on your computer.
   - Select the correct board (Arduino Uno) and port in Mixly's settings.
2. **Create the Program Using Blocks:**

To blink an LED, follow these steps:

- Drag the "Control" block that says "Set digital pin" from the Control category.
  - Set this to pin 13 and choose HIGH (this will turn the LED on).
- Drag the "Time" block and choose the "Delay" option.
  - Set the delay to 1000 milliseconds (1 second).
- Drag another "Control" block to set digital pin 13 to LOW (this will turn the LED off).
- Drag another "Time" block and choose "Delay".

- Set this delay to 1000 milliseconds as well (this will keep the LED off for 1 second).

This combination of blocks will create a loop where the LED is turned on for 1 second, then turned off for 1 second, creating the blinking effect.

3. **Loop the Program:**
   - Add a "Forever" loop from the Control category to ensure the LED continues blinking indefinitely.
   - Place all the blocks inside the "Forever" loop.

Step 3: Upload the Program to the Arduino

1. Check Your Connections:
   - Ensure that your LED and Arduino are connected properly.
2. Upload the Code:
   - Once your block-based program is complete, click the Upload button in Mixly.
   - Wait for the program to compile and upload to the Arduino.

Step 4: Observe the Blinking LED

- After uploading, the LED connected to pin 13 should start blinking with a 1-second interval (on for 1 second, off for 1 second).
- You have successfully made an LED blink using the Mixly block programming environment!

Explanation of the Program:

- Pin 13 HIGH: Turns the LED on by sending a voltage signal to pin 13.
- Delay (1000ms): Keeps the LED on for 1 second.
- Pin 13 LOW: Turns the LED off by removing the voltage signal.
- Delay (1000ms): Keeps the LED off for 1 second.

This process repeats in a loop, creating the blinking effect.

Challenge:

- Can you change the delay to make the LED blink faster or slower?
- Try using a different pin or connecting multiple LEDs and controlling them separately.

This activity introduces you to basic circuit control and programming with Mixly, which is useful for more complex projects involving sensors, motors, and other electronic components.